

Sequence Diagram – Borrow Equipment Request

If the image is not clear, the detailed behavior of the sequence diagram is described as follows.

Participants

- Student – person who wants to borrow a device
- UI – system interface used by the student
- Borrower – application logic that handles user input
- BorrowRequest – component that manages borrow requests
- Staff – approver who reviews the request
- Equipment – component that stores device availability
- Notification – component that sends messages to users

Main Scenario

1. The Student uses the UI to send a *request to borrow a device*.
2. The UI forwards the entered information (device list, purpose, time period, quantity, etc.) to the Borrower component.
3. The Borrower creates a new request by invoking an operation on BorrowRequest.
4. BorrowRequest queries the Equipment component to check whether the requested device(s) are available for the specified period.
5. The Equipment component replies with the availability result.

Alternative Fragment – Device Availability

- [Device Available]
 6. BorrowRequest sets the request status to Pending.
 7. BorrowRequest informs the Borrower/UI that the request has been recorded and is waiting for approval.
- [Device Not Available]
 - 6'. BorrowRequest sends an error message back to the Borrower/UI indicating that the selected device is not available.
 - 7'. The UI displays the unavailability message to the Student, who may correct the request or choose another device.

Approval Process

8. The Staff accesses the system and calls an operation on BorrowRequest to *review* the pending request.

[Approve Request]

9. If the staff decides to approve, they trigger an approveRequest() operation on BorrowRequest.
10. BorrowRequest updates its internal status to Approved.
11. The updated status is sent back to the UI, which confirms to the Student that the request has been approved.
12. The UI then calls the Notification component to send an approval message (e.g., email or in-app notification) to the student.

[Reject Request]

- 9'. If the staff rejects the request, they invoke rejectRequest() on BorrowRequest.
- 10'. BorrowRequest changes the status to Rejected and may store the rejection reason.
- 11'. The new status is returned to the UI, which shows the rejection result to the Student.
- 12'. The UI also uses the Notification component to send a rejection message to the student.

